Bayesian Optimisation of a Bio-Based Biodegradable Polyester Thermoset

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Abstract

Glass reinforced plastic composites are ubiquitous in the wind turbine manufacturing industry, and are commonly manufactured using general-purpose unsaturated polyester resin thermosets. Unfortunately, these resins are derived from oil-derived aromatic monomers making them both unsustainably sourced and hard to recycle. Bio-based and biodegradable alternatives exist but have not seen widespread commercial success; this paper seeks to address the question of whether bio-based branching thermoset polyesters can be optimised using Bayesian optimisation such that desirable properties are improved upon. This is achieved by identifying a bio-based and biodegradable polymer, poly(glycerol citrate itaconate), identifying a desirable objective variable and some tunable predictor variables, and finally using Bayesian optimisation to sample-efficiently identify well-performing samples. The most salient findings included the Bayesian optimisation routines successfully optimising the proxy variable in a sample efficient fashion, improving upon common deterministic techniques such as grid sampling.

1 Introduction

In 1987, the concept of 'sustainable development' was defined by the United Nations (UN) Brundtland Commission as a development which seeks to "meet the needs and aspirations of the present without compromising the ability to meet those of the future" [29]. Wind turbine manufacturing is an industry shaped by this ideal, with its end-products contributing towards the UN's 13th sustainable development goal (SDG) which urges "urgent action to combat climate change and its impacts" [27]. Unfortunately, wind turbine blades are commonly manufactured from a composite comprised of thermosetting polymer matrix and reinforcing material, these strong thermoset matrices are chemically cross-linked which makes them difficult to reprocess or remould [17]. Given this resistance to recycling and a relatively common rate of replacement (ca. every fifteen years in China [8]) this industry generates a large volume of practically unrecyclable thermoset waste which contravenes the UN's 12th SDG to "ensure sustainable consumption and production" [27]. In addressing the industry's sustainability shortcoming identified above, the literature forks into two groups: firstly, papers which focus upon improving the recyclability of contemporary materials, whether by thermal, chemical, mechanical or high-voltage fragmentation methods [28, 5, 9]. Secondly, there are papers aimed at the development of properties in future replacement materials that would facilitate recycling [16], such as introduction of chemically cleavable bonds into the thermoset polymers used [25]. This paper's research sought to ascertain whether Bayesian optimisation could be used to sample efficiently improve a desirable property in a novel biodegradable bio-based thermoset.

The composite matrix used for wind turbine blades is comprised of a moulding thermoset, with the most prevalently manufactured pre-thermoset form being general-purpose unsaturated polyester resin (GP-UPR) [6]. These GP-UPRs are based on a mixture of reactive diluent (e.g. styrene)

and linear aromatic unsaturated polyester (e.g. poly(propylene glycol maleate phthalate)), which are cured chemically or thermally to generate the final thermosetting matrix [1]. Although polyesters are known to exhibit biodegradability (due to their labile ester bonds being prone to both biotic and abiotic hydrolysis [22]) it is the case that polyesters with aromatic moieties are more resistant than aliphatic counterparts to such degradation [19]. Consequently, thermosets derived from GP-UPRs synthesised from the aromatic monomer phthalic acid [1] can be reasonably expected to exhibit a high resistance to biodegradation; which was part of their original allure as a material. An archetypical example of an aliphatic polyester would be poly(glycerol sebacate), which uses the monomers glycerol (triol) and sebacic acid (dicarboxylic acid) to form a branching polyester thermoset on thermal curing via polyesterification [31]. Given their natural biodegradability, this research chose to focus upon aliphatic polyesters.

Another concern associated with GP-UPRs with respect to the UN's 12th SDG is that of a material's provenance; if the underlying chemistry of a polymer is petrochemical in origin, then any carbon liberated from the material at end-of-life (whether by combustion for energy or biodegradation in landfill) will release fossil carbon from the long-term carbon cycle into the short-term carbon cycle that exists above the endogenic reservoir, contributing via an anthropogenic route towards global climate change. Since the aforementioned GP-UPRs are synthesised from the oil-derived feedstock monomers propylene glycol, maleic acid, and phthalic acid, the resultant thermosets contain endogenic (fossil-derived) carbon [15]. This research avoided these feedstocks and centred on a bio-based alternative, with inspiration drawn from a chemical system presented in the literature by Alberts, where a combination of glycerol and citric acid were cured to form a bio-based branched aliphatic polyester thermoset [2]. We build upon this research by adopting the chemical system and adapting it slightly to include a third monomer, itaconic acid, which provides a second carboxylic acid with a lower degree of functionality (dicarboxylic acid). Notably, Alberts found that physical properties of the end thermoset poly(glycerol citratea) could be tuned by modifying the monomeric stoichiometries of the prepolymer resin synthesised [2]. Given this, we identified mass loss as a variable of interest during the synthesis of poly(glycerol citrate itaconate) as it was a proxy for polymerised content within a sample, and sought to optimise this by tuning the stoichiometries of the prepolymer resins mixed.

Poly(glycerol citrate itaconate) is formed from its monomers by polyesterification, a process which is dependent upon multiple variables such as temperature, stoichiometric ratios, and additive presence [2]. Additionally, reaction times for rigid solid thermoset materials are on the order of weeks when uncatalyzed [26] (catalysis commonly involves strong fuming acids which corrode equipment and are best avoided [23]). Given the high dimensionality and temporally expensive function evaluation exhibited by this family of polyesters, a sample efficient global optimisation technique by the name of Bayesian optimisation was elected as a suitable means of optimisation. Alternative global optimisation techniques were considered but ruled out; the hill-climbing class gradient descent algorithm was reliant on derivatives of the objective function [24], which would have been uncertain and practically difficult to obtain given the noisy real-world nature of the black-box function evaluated. The population class evolutionary and particle swarm optimisation algorithms were suitable but avoided in favour of the more sample efficient surrogate class of algorithms. Surrogate class algorithms are variously formulated, but the three major forms are distinguished by the surrogate models they leverage- polynomial response surface model , kriging (aka gaussian process re-

gression), radial basis function [21]. Kriging is preferable to response surface modelling since it makes no assumption of polynomial order, avoiding a priori bias in surrogate models generated. Furthermore, radial basis function based surrogate models provide no estimates of uncertainty around the fitted function; functions and associated uncertainties derived using kriging provides the means by which bayesian optimisation effectively balances exploitation and exploration of the parameter space. This research therewith saw the development of a part-computational and part-laboratory based workflow for the optimisation of mass loss in poly(glycerol citrate itaconate) leveraging Bayesian optimisation predicated on kriging.

Originally coined in the 1970s by Mockus et al. [18], Bayesian optimisation was later developed further in the 1990s by Jones et al. into a technique called the efficient global optimisation algorithm [14]- this name and technique was thereafter adopted and popularised by the DoE community. More recently, the original name has seen a revival as the technique and a plethora of adaptations have been re-discovered and implemented in a range of contemporary communities including robotics [7], bioengineering [3], and machine learning [10]. In materials science, Bayesian optimisation has been used to predict crystalline structures [30], carry out sample-efficient optimisation of density functional theory simulations [4], and multi-objective optimisation of stability and ion-conductivity in materials

2 Method

In optimising the properties of a poly(glycerol citrate itaconate) thermoset, interlinking method was developed spanning both practical experimental technique and mathematical framing. Given this method, a computer program was created providing a command-line interface facilitating human-in-the-loop optimisation.

2.1 Chemical Background

The condensation reaction occurring between carboxyl and hydroxyl groups is an esterification reaction (See Eq. 1).

$$RCOOH + HOR' \rightarrow RCOOR' + HOH \tag{1}$$

It is described as a condensation reaction since a small by-product molecule, water, is generated as each ester bond is formed. As opposed to simple esterification, the desired polyester thermoset, poly(glycerol citrate itaconate), is the product of polyesterification between multi-functional monomers.

Polyesterification is a form of polycondensation reaction which takes place as follows: an initial ester is formed from two difunctional monomers, where one must at least be a diol and the other a diacid (alternatively, each may possess a carboxyl and a hydroxyl group), producing a molecule of water and an ester with two spare functional groups- these groups can subsequently react with more difunctional monomers causing polymerisation into a linear polyester (See Fig. 1). When monomers with a functionality greater than two are introduced into a polyesterification reaction, then side chains can form along the primary backbone of the growing polyester, this causes the development of a branching polymeric structure (See Fig. 2). When a side chain reaches another large growing polyester molecule and reacts with it a crosslink is said to have developed; when widespread crosslinking occurs generating a 3D network of polyester molecules a thermoset polyester has been created. Figure 2 shows how the monomers employed in the synthesis of poly(glycerol citrate itaconate) give rise to this branching thermoset structure.

Importantly, during the polyesterification process water is generated as the only by-product, it is therewith assumed that virtually all mass loss between prepolymer and final cured thermoset would be associated with water evaporation. Given that each

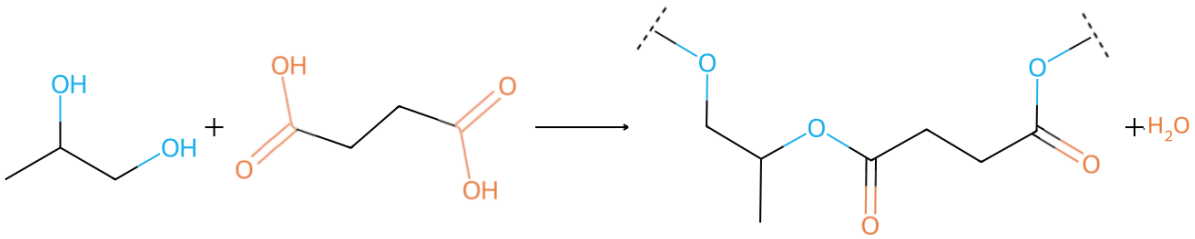

Figure 1: ***SimpleLinearEsterification****Archetypical polyesterification between a diol (propylene glycol) and a diacid (succinic acid) forming a linear polyester. Functional groups participating in the reaction are highlighted: hydroxyl groups blue, and carboxyl groups orange.*

molecule of water produced corresponds to an ester bond formed, mass loss can be said to be a loose proxy for extent of polymerisation within a sample [26]. This research identifies this as a metric of interest and sought to maximise the mass loss during optimisation of poly(glycerol citrate itaconate) synthesis: increasing the extent of polymerisation and possibly improving the degree of cross-linking and associated strength of these bio-based thermosets.

2.2 Experimental Method

Synthesis of the poly(glycerol citrate itaconate) thermoset was achieved via an uncatalysed polyesterification reaction (See Figure 2). Acid catalysis was eschewed to avoid corrosion damage to equipment which slowed but failed to halt the polyesterification process [4]. The monomers glycerol (99+%), citric acid (99.6%), and itaconic acid (99+%) used were supplied by *Fisher Scientific International* and manufactured by *Acros Organics*.

Thermoset synthesis was two-step: firstly, a homogenous prepolymer resin was prepared by stirring solid monomers (acids) together with liquid monomer (glycerol) in-vitro atop a hotplate and raising the temperature (maximum 140°C) until the solids were completely dissolved. Note, 140°C was adopted to accelerate dissolution without encouraging decarboxylation of the citric acid to itaconic acid [2]. Secondly, prepolymer was cured into the desired thermoset by transferral to a silicone mould before being exposed to elevated temperatures (90° C) for an extended period (up to two weeks) within a drying oven.

Preliminary experimentation showed that temperatures exceeding 100° C caused boiling of byproduct water within samples. Gaseous water from boiling was entrapped within the curing polyester creating vesicles within the final structure, resulting in unpredictable densities of the final material. Given both the random and uncontrollable nature of samples generated in this fashion, drying oven temperatures were held below 90° C to avoid this imponderables affect upon the experiments.

As seen in Equation 2, determination of mass loss associated with the synthesis of a thermoset sample (Δm) was obtained by measuring the difference between both the initial mass of prepolymer resin (m_p) and the final mass of cured thermoset (m_t). Measurements were made with a mass balance accurate to 0.01 g $\pm$ 0.01 g; since prepolymer samples weighed 3 g and mass loss was confined to between 0.3-0.9 g we assumed systematic errors accrued through uncertainty in mass balance readings to be negligible.

$$\Delta m = m_p - m_t \tag{2}$$

Equation 3 shows how sample mass loss (Δm) is normalised into a percentage format ($\Delta m_{\%}$), this allowed inter-comparison since individual samples had slightly varying initial prepolymer resin masses (m_p).

$$\Delta m_{\%} = \frac{\Delta m \times 100}{m_p} \tag{3}$$

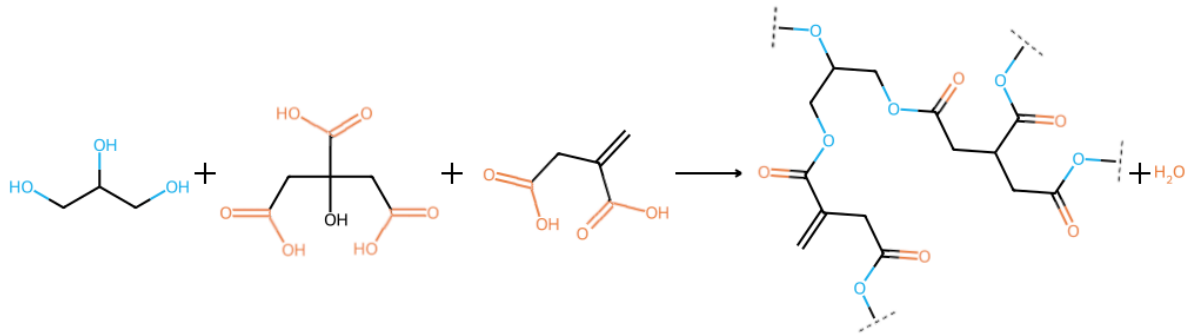

Figure 2: ***SimpleReactionOverview****Polyesterification between glycerol, citric acid, and itaconic acid forming poly(glycerol citrate itaconate). Functional groups participating in the reaction are highlighted: hydroxyl groups blue, and carboxyl groups orange.*

Residence time within the drying oven for a sample was controlled by periodic measurement of sample mass; mass was plotted against time, an exponential decay curve fitted, and the sample removed from the oven once an arbitrary gradient was reached.

2.3 Optimisation Framing

Having selected percentage mass loss within a thermoset sample ($\Delta m_{\%}$) as a suitable target variable for optimisation, predictor variables were defined and constrained creating a suitable parameter space describing a full range of prepolymer monomeric compositions.

The polyesterification of monomers to polymer seen in Figure 2 is presented in Equation 4 as an unbalanced stoichiometric equation linking monomeric reagents glycerol (G), itaconic acid (It), and citric acid (Ci) with the product polymer poly(glycerol citrate itaconate) (P(GCiIt)) and byproduct water. Equation 4 introduces two stoichiometric variables representing the moles of glycerol (n_G), citric acid (n_{Ci}) and itaconic acid (n_{It}), and a variable representing the unknown number of moles of water produced (v).

$$n_G\ G + n_{Ci}\ Ci + n_{It}\ It \rightarrow 1\ P(GCiIt) + v\ H_2O \tag{4}$$

The polyesterification reaction can be considered to consist of two possible polymerisation reactions between the triol glycerol and either the triacid citric acid or the diacid itaconic acid. By making a constant of the glycerol stoichiometric variable ($n_G = 1$), Figure 3 shows how the two separate reactions can be considered as if they were stoichiometric ranges, where there exist stoichiometric bounds between which polymerisation is possible.

These stoichiometric bounds are used to directly constrain the stoichiometric variables; Equation 5 describes the lower (n^l_{Ci}) and upper (n^u_{Ci}) bounds placed on the citric acid stoichiometric variable, whilst Equation 6 shows those for itaconic acids lower (n^l_{It}) and upper (n^u_{It}) bounds.

$$n^l_{Ci} \leq n_{Ci} \leq n^u_{Ci} = 0.5 \leq n_{Ci} \leq 2.0 \tag{5}$$

$$n^l_{It} \leq n_{It} \leq n^u_{It} = 1.0 \leq n_{It} \leq 2.0 \tag{6}$$

In addition to the basic polymerisable bounds imposed on the two stoichiometric variables, a further constraint must be imposed for a successful optimisation: it must be possible for permutations of prepolymer monomeric mixtures to allow one acid to entirely marginalise the other whilst ensuring that a prepolymer will still polymerise- this constraint ensures that any polymerisable stoichiometric mixture of monomers is possible. To achieve this, a three-dimensional bounded domain (a.k.a parameter space, A) was devised consisting of three fundamental predictor variables: firstly, a citric acid strength predictor variable (s_1) influencing

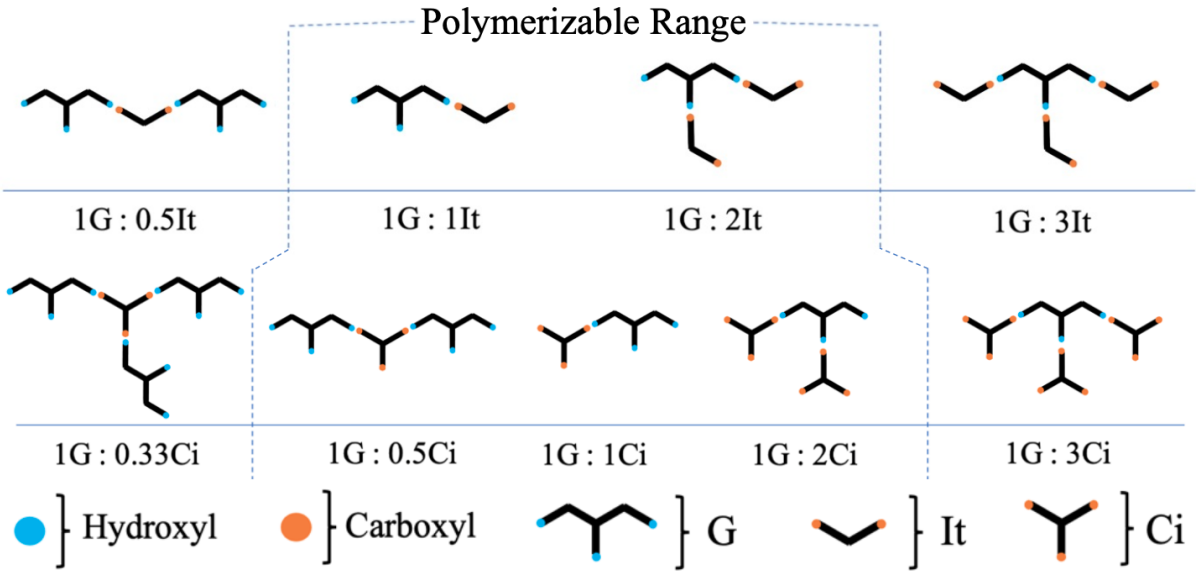

Figure 3: ***PolymerisableRange****Simplified chemical representations displaying the polymerisable range of the two different triol-acid combinations encountered during poly(glycerol citrate itaconate) synthesis. In the legend: "G" glycerol, "It" itaconic acid, "Ci" citric acid.*

the citric acid stoichiometric variable. Secondly, an itaconic acid strength predictor variable (s_2) influencing the itaconic acid stoichiometric variable. Thirdly, a balancing predictor variable (b_1) capable of trading off the two acid systems, influencing both the citric and itaconic acid stoichiometric variables. Equation 7 describes the parameter space A as a multi-dimensional real of d dimensions, where $d = 3$ given our three predictor variables.

$$A \subseteq \mathbb{R}^d \tag{7}$$

All three predictor variables took constrained ranges of between 0 and 1 ($s_1 = [0, 1], s_2 = [0, 1], b_1 = [0, 1]$), and were linked mathematically to the stoichiometric variables (See Equations eqs. (8) and (9)).

$$n_{Ci} = (n^l_{Ci} + (n^u_{Ci} - n^l_{Ci}) \times s_1) \times b_1 \tag{8}$$

$$n_{It} = (n^l_{It} + (n^u_{It} - n^l_{It}) \times s_2) \times (1 - b_1) \tag{9}$$

In summation, any random set of predictor variable values drawn from their respective constrained ranges of 0 to 1 (e.g. $s_1 = 0.75, s_2 = 0.25, b_1 = 0.5$) could be used to calculate a unique permutation of stoichiometric variables (e.g. $N_{Ci} = 0.81, N_{It} = 0.63$) which, on mixing of a prepolymer in accordance with Equation 10 and curing, would give rise to a unique polymer and associated mass loss (e.g. $\Delta m_{\%} = 30$).

$$\begin{aligned} n_G\ G + 0 \leq n_{Ci} \leq & n^u_{Ci}\ Ci + \\ & 0 \leq n_{It} \leq n^u_{It}\ It \rightarrow \\ & 1\ P(GCiIt) + v\ H_2O \end{aligned} \tag{10}$$

The sequence of calculations linking abstract predictor variables (s_1, s_2, b_1) to the practical stoichiometric variables (n_{Ci}, n_{It}) and onwards to the target variable of mass loss ($\Delta m_{\%}$), allows the expression of a black-box mapping between predictors and targets. A black-box mapping is described by Equation 11, where an unknown function (f) maps any point in our parameter space (A) to a single real number value ($\mathbb{R}$).

$$f : A \rightarrow \mathbb{R} \tag{11}$$

Equation 12 more explicitly expresses the black-box function in terms of the our target variable and predictor variables at a given sample number (i). Note that sample number is an integer counter used to track the number of function evaluations carried out during a sequential optimisation.

$$\Delta m_{\%} = f(\mathbf{x}) = f([s^i_1, s^i_2, b^i_1]) \tag{12}$$

Our optimisation problem is expressed in Equation 13 as the maximisation of the black-box function, where the global optimum set of predictor variables ($\mathbf{x}^*$) is sought.

$$\mathbf{x}^* = \arg\max_{\mathbf{x} \in A} f(\mathbf{x}) \tag{13}$$

2.4 Optimisation Algorithm

The sequential optimisation algorithm employed during this work is summarised in Figure 4. Cessation of the optimisation is controlled by an if-else clause comparing an initially specified permissable number of function evaluations (a.k.a. sample limit, p_{tot}) with the current sample number (i): when $i < p_{tot}$ the algorithm continues. A second if-else statement controls whether a dataset exists from which a Gaussian process prior might be built for Bayesian sampling, when available, the algorithm uses the Gaussian process to draw a pre-defined number of samples (p_{iter}) from the parameter space for evaluation. If the prior-forming dataset is lacking, an initial quasirandom (Sobol) sampling of the parameter space with a pre-defined number of deterministic samples (p_{init}) is initiated. Samples drawn from the parameter space are processed into stoichiometric variables (n_{Ci}, n_{It}), with which an experimenter can manufacture sample(s) before evaluating the target variables $\Delta m_{\%}$ value(s) and returning these to the algorithm.

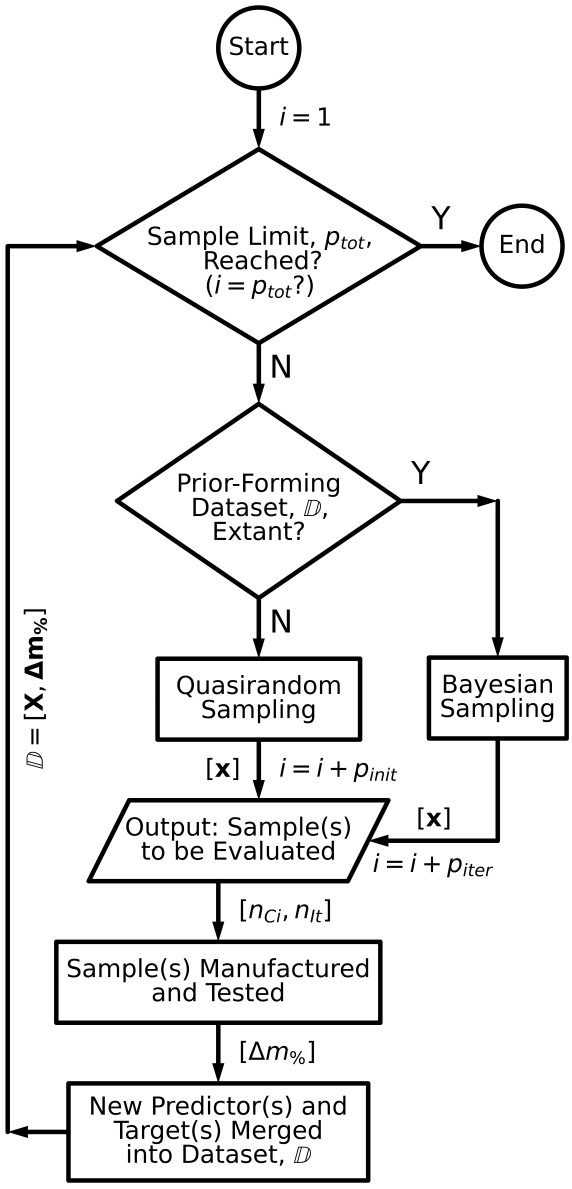

Figure 4: ***OptimisationAlgorithm*** *Flow diagram depicting the sequential optimisation algorithm developed.*

On calculation of stoichiometric variables during an iteration, the algorithm merges all new sets of predictors variable values into a growing predictor-matrix ($\mathbf{X}$), shown in Equation 14.

$$\begin{aligned}\mathbf{X} =& [\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, ...] \\ =& [[s_1^{i=1}, s_2^{i=1}, b_1^{i=1}], [s_1^{i=2}, ...], ...]\end{aligned} \tag{14}$$

On conclusion of an iteration, the algorithm merges all data associated with samples evaluated into an optimisation-governing dataset. Equation 15 presents this dataset ($\mathbb{D}$) as a matrix which consists of the

predictor-matrix ($\mathbf{X}$) and an array of target variable values ($\mathbf{\Delta m}_{\%}$).

$$\begin{aligned}\mathbb{D} =&[\mathbf{X}, f(\mathbf{X})]\\ =&[\mathbf{X}, \mathbf{\Delta m}_{\%}]\\ =&[[\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, ...], [\Delta m_{\%}^{i=1}, ...]]\end{aligned} \tag{15}$$

2.5 Sampling Procedures

During a sequential optimisation two forms of sampling were employed: firstly, a deterministic space-filling sampling of a quasirandom nature was used to representatively sample the parameter space when initialising the optimisation. Subequently, a Bayesian sampling technique was used to sample-efficiently guide the optimisation's search for the global optimimum x^*.

2.5.1 Initial Quasirandom Sampling

Optimisation routines were initiated by representatively sampling across the parameter space A; achieved using the quasirandom Sobol sampling algorithm to draw a preset number (p_{init}) of samples from across A. Sobol sampling was employed as it provides the same or better space-filling properties across $\mathbb{R}^{1\leq d\leq 40}$ domains as compared with pseudorandom or grid techniques [13]. On evaluation of the initial representative set of samples ($[\mathbf{x}^{i=1}, ..., \mathbf{x}^{i=p_{init}}]$) for the target variable values ($[\Delta m_{\%}^{i=1}, ..., \Delta m_{\%}^{i=p_{init}}]$), an initial prior-forming dataset ($\mathbb{D}^{i=p_{init}}$) is produced. This research employed only two quasirandomly sampled initial prior-forming datasets, facilitating inter-comparison of various Bayesian optimisation routines attempted; the smaller ($\mathbb{D}^{i=8}$) and larger ($\mathbb{D}^{i=16}$) of these are visualised in Figure 5.

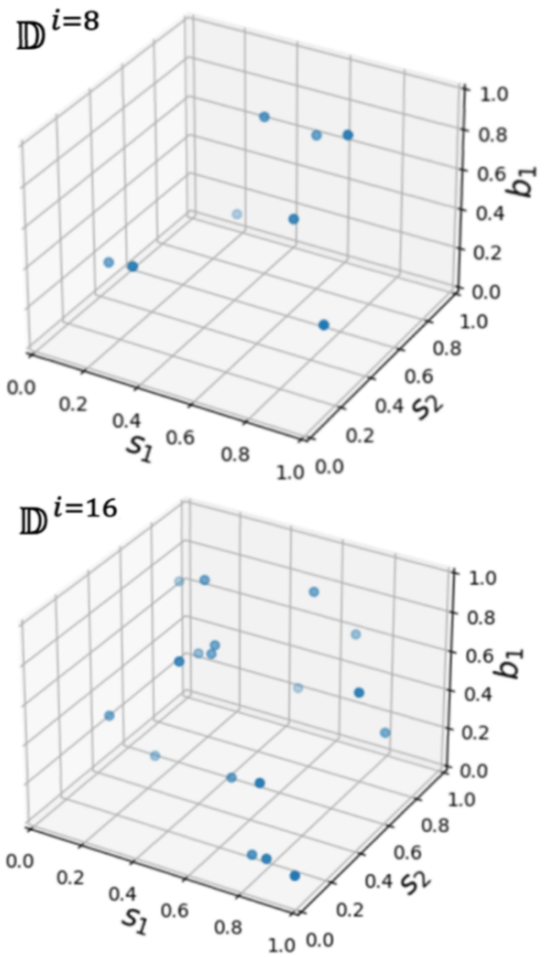

Figure 5: ***BayesianPriors*** *These 3D graphs depict the two quasirandom samplings of the parameter space.*

2.5.2 Sequential Bayesian Sampling

Sequential sampling used in this research abided by Bayes theorem (see Equation 16), where a posterior probability $P(H|E)$ (updated probability of hypothesis H given new evidence E) is equal to the prior probability (original probability of hypothesis) multiplied by the likelihood $P(E|H)$ (probability of evidence given hypothesis) divided by the marginal probability $P(E)$ (probability of evidence). As new evidence becomes available, the posterior probability is recycled into the equation as a prior alongside the new evidence and updated to a new posterior [12].

$$\overset{\text{posterior}}{P(H|E)} = \frac{\overset{\text{prior}}{P(H)} \times \overset{\text{likelihood}}{P(E|H)}}{\underset{\text{marginal}}{P(E)}} \tag{16}$$

Bayesian optimisation uses a Bayesian statistical model called a Gaussian process regression model (a form of Bayesian posterior distribution) derived from Gaussian process regression of a dataset $\mathbb{D}$ [20]. Equation 17 shows how a gaussian process, $\mathcal{GP}$, is a distribution over functions which is completely described by both a mean function, m, and a covariance function, k.

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \tag{17}$$

More intuitively, Equation 18 describes the gaussian process as a surrogate function $\hat{f}(\cdot)$ describing the link between a candidate array of predictor values $\mathbf{x}$ and a tuple containing the mean and the variance of a normal distribution over values of f at x.

$$\hat{f}(\cdot) : \mathbf{x} \mapsto [m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')] \tag{18}$$

As an infinite-dimensional generalisation of multivariate normal distributions the gaussian process inherits the properties of a conjugate prior [11], allowing the initial posterior (e.g. $\mathcal{GP}^{i=8}$) to be recycled as a prior alongside new data (e.g. $\mathbb{D} + [\mathbf{x}^{i=9}, \Delta m^{i=9}]$) to generate an updated posterior (e.g. $\mathcal{GP}^{i=9}$) in accordance with Bayes theorem (see 16). The surrogate model therewith evolves during the course of sequential optimisation as the number of iterations, i, increases: corresponding to the updating of beliefs when considering new evidence which hallmarks all Bayesian techniques.

2.6 Benchmarks

3 Results

4 Discussion

5 Conclusion

6 Bibliography

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